

PROBLEM WITH V_z DISTRIBUTION

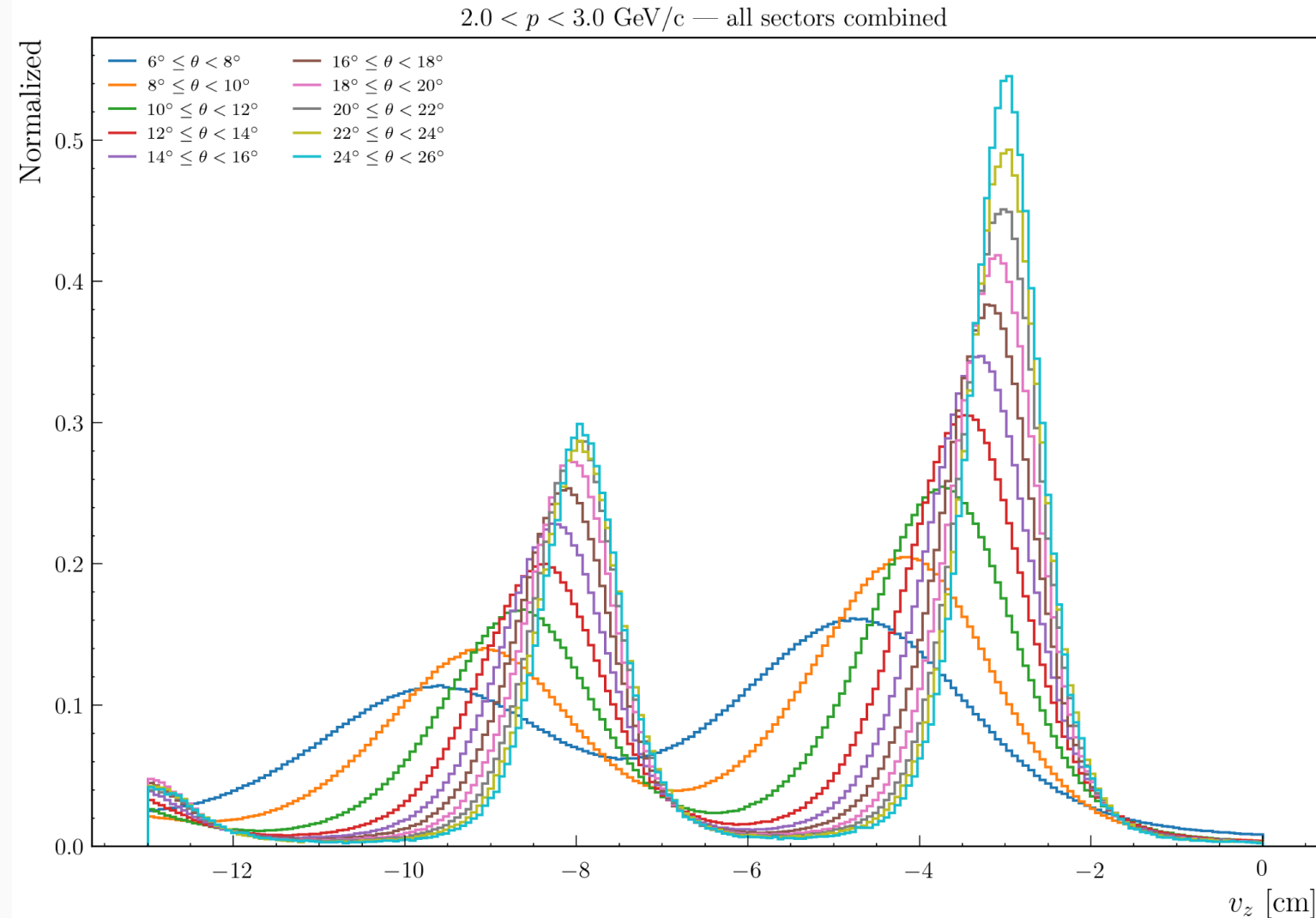
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06/18/26



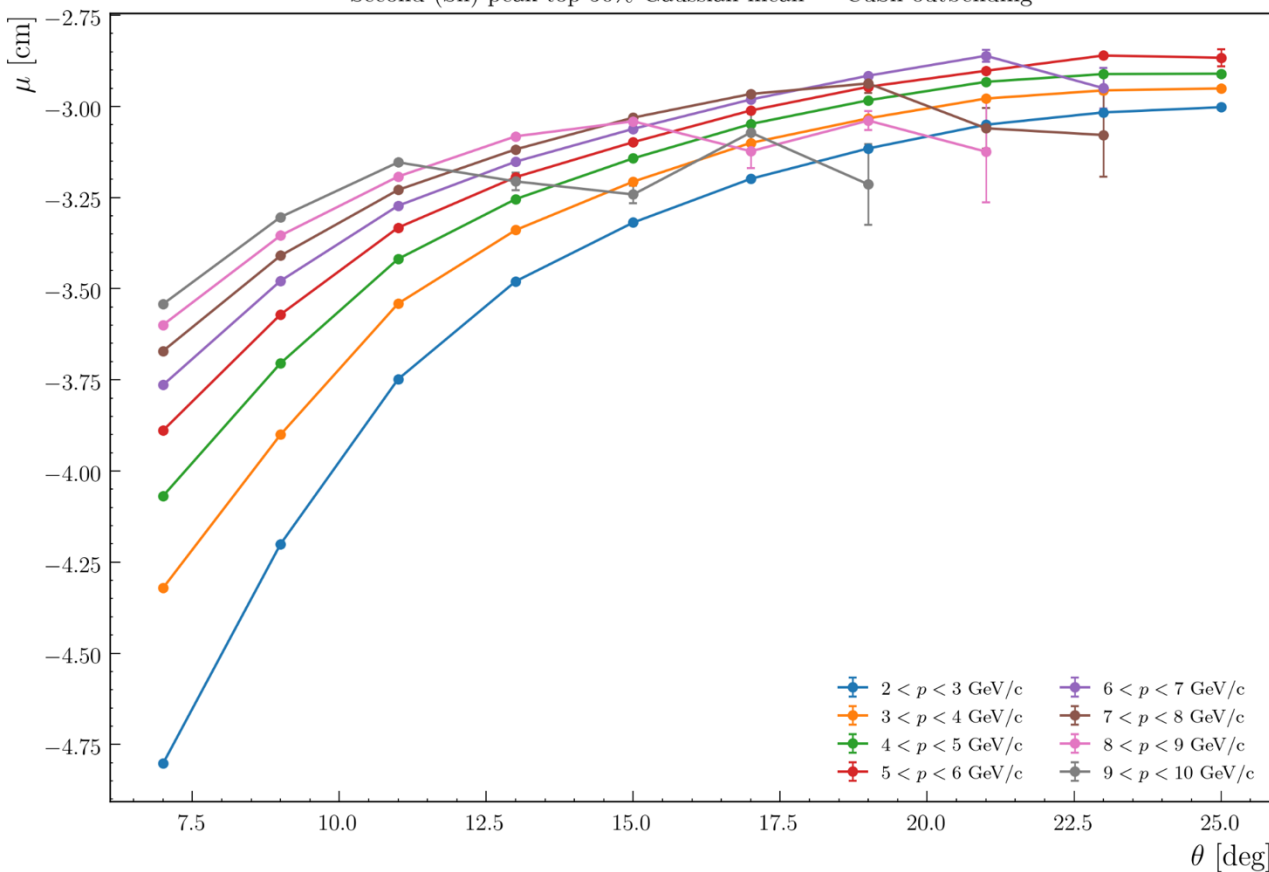
PROBLEM WITH V_z DISTRIBUTION

- I wanted to develop V_z cuts for RG-D. We have a specific target setup:
 - Two fine foil ($\sim 100\mu\text{m}$) separate by 5cm, and positioned around -5cm
 - Found that the position of the peaks shift and the peaks become also broader as inverse of momentum and θ
 - Here it is for electron (with usual cuts)

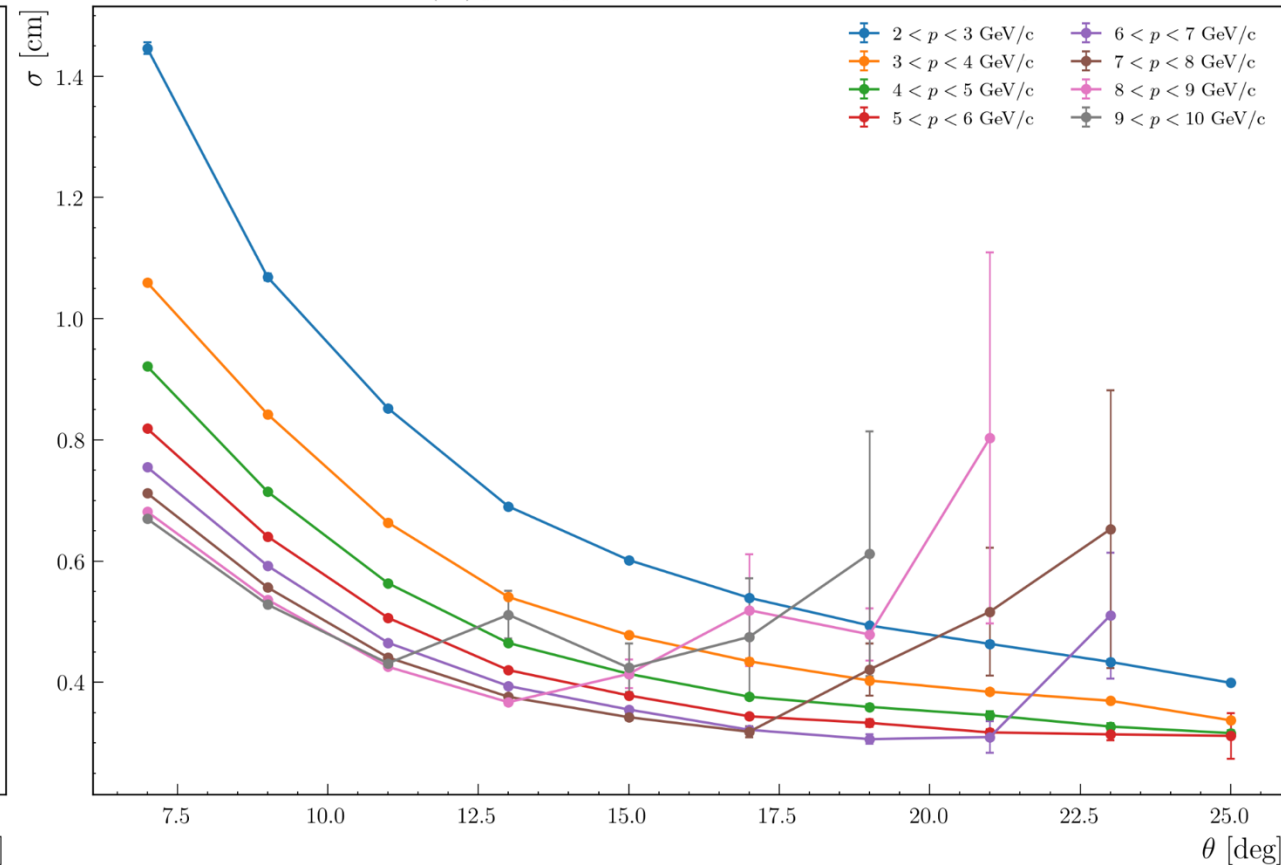


PROBLEM WITH V_z DISTRIBUTION

Second (Sn) peak top-50% Gaussian mean — CuSn outbending

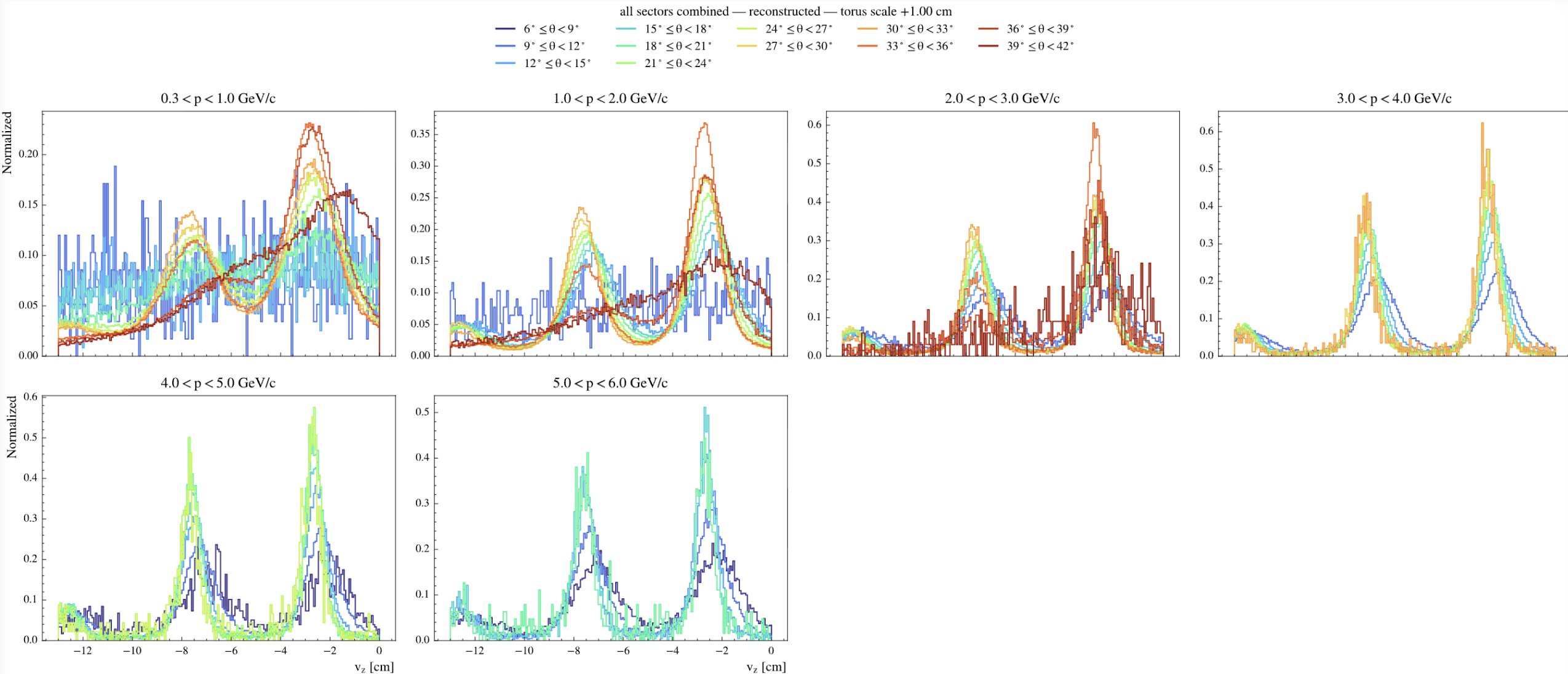


Second (Sn) peak top-50% Gaussian width — CuSn outbending



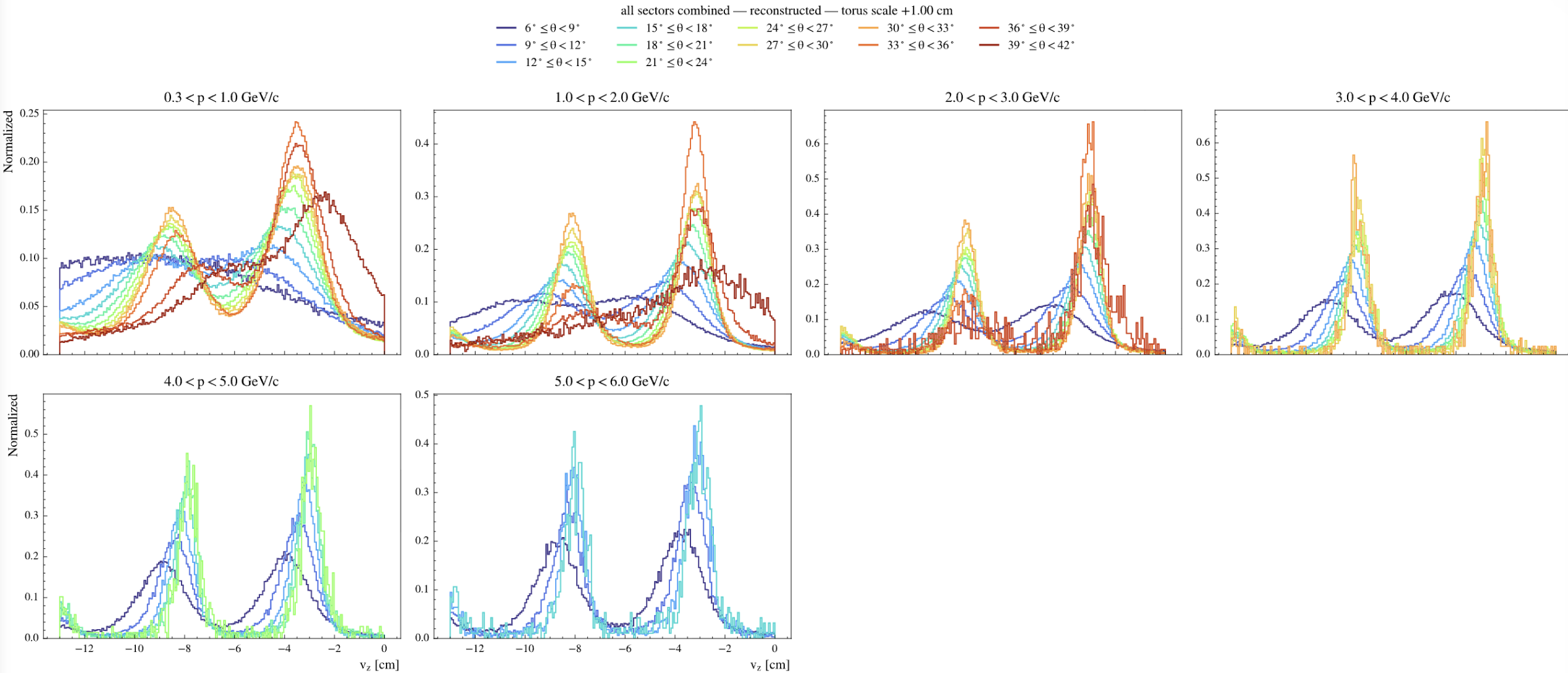
PROBLEM WITH V_z DISTRIBUTION

- Similar effect for π^+ :



PROBLEM WITH V_z DISTRIBUTION

- Similar effect for π^- :



PROBLEM WITH VZ DISTRIBUTION

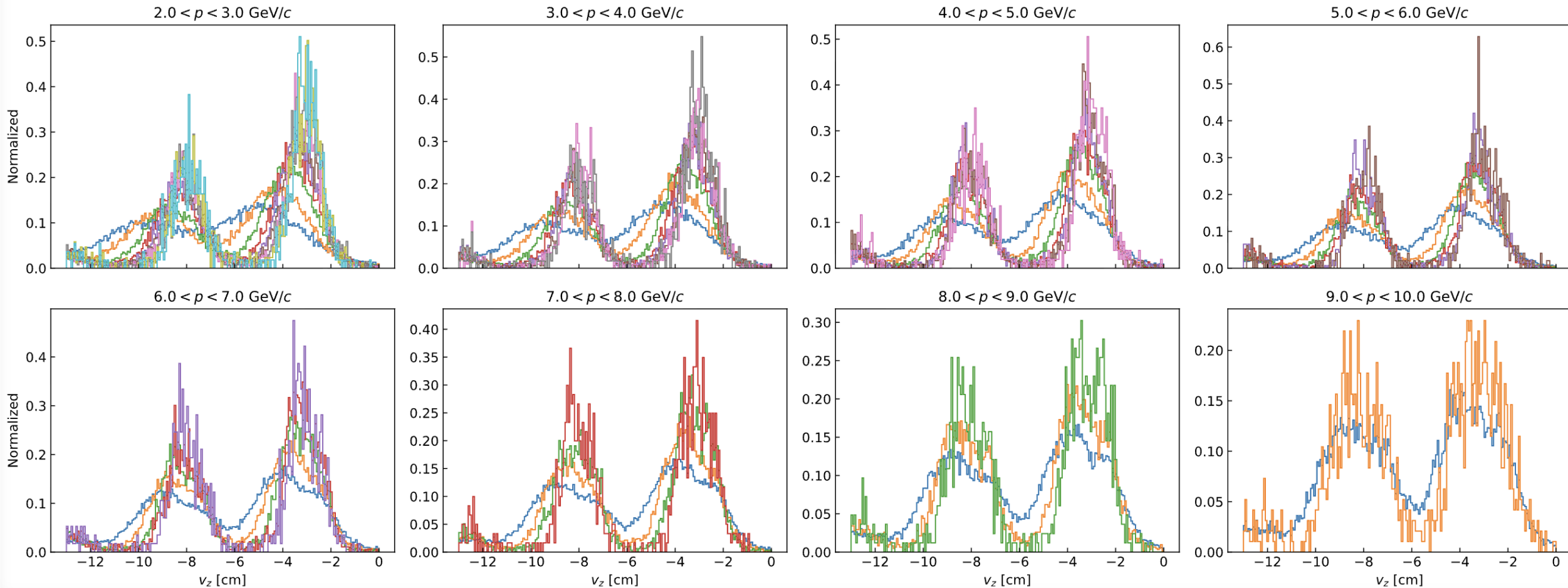
- Try different modifications of the magnetic field map to correct for that:
 - To test that, I am using the field maps downloaded from clasweb:
 - Full_torus_r501_phi361_z501_31Mar2021.dat, Symm_solenoid_r601_phi1_z1201_21May2019.dat
 - Then I need to swim the particle in the sum of the two magnetic field. For that:
 - I use the the position and momentum in DC region 3 as starting point
 - Then use ODE solver (adaptive Dormand–Prince 5(4)) to solve equation of motion for charged particle in magnetic field
 - And swim back the particle until the DOCA to the beamline
- Tested:
 - Shift along x, y and z axis for the solenoid and the torus (one axis and one magnet maps at a time)
 - Didn't fix our problem
 - Try to change the strength/scale of the torus field

PROBLEM WITH V_z DISTRIBUTION

- Try to change the strength/scale of the torus field: torus scale +1

e^- , all sectors combined — swum (DC) — torus scale -1.00 cm

— $6^\circ \leq \theta < 8^\circ$	— $10^\circ \leq \theta < 12^\circ$	— $14^\circ \leq \theta < 16^\circ$	— $18^\circ \leq \theta < 20^\circ$	— $22^\circ \leq \theta < 24^\circ$
— $8^\circ \leq \theta < 10^\circ$	— $12^\circ \leq \theta < 14^\circ$	— $16^\circ \leq \theta < 18^\circ$	— $20^\circ \leq \theta < 22^\circ$	— $24^\circ \leq \theta < 26^\circ$

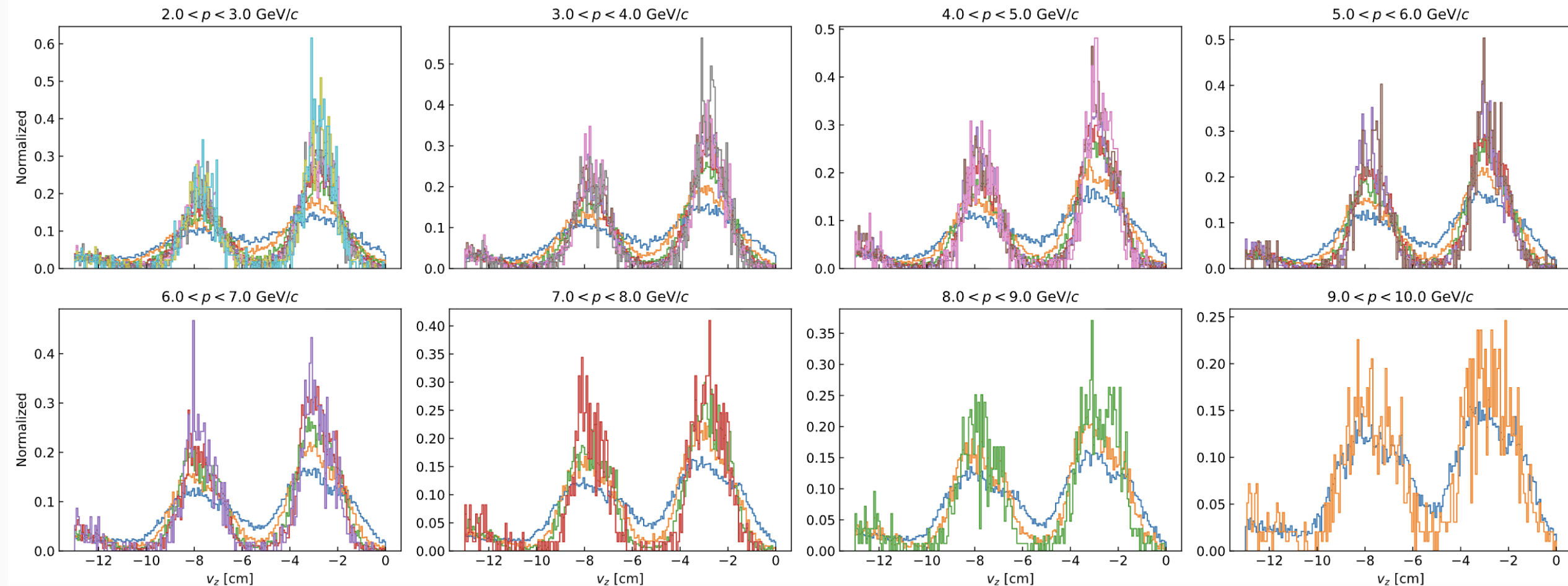


PROBLEM WITH V_z DISTRIBUTION

- Try to change the strength/scale of the torus field: torus scale +0.998

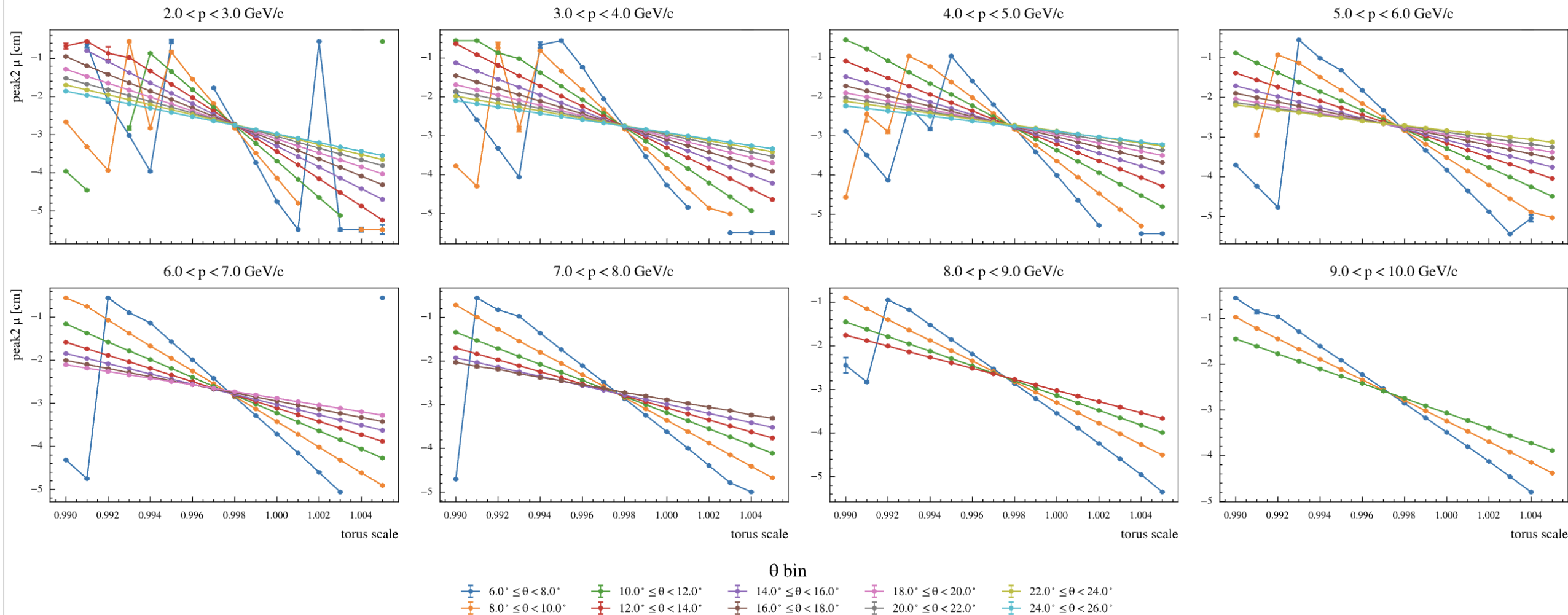
e^- , all sectors combined — swum (DC) — torus scale -1.00 cm

— $6^\circ \leq \theta < 8^\circ$	— $10^\circ \leq \theta < 12^\circ$	— $14^\circ \leq \theta < 16^\circ$	— $18^\circ \leq \theta < 20^\circ$	— $22^\circ \leq \theta < 24^\circ$
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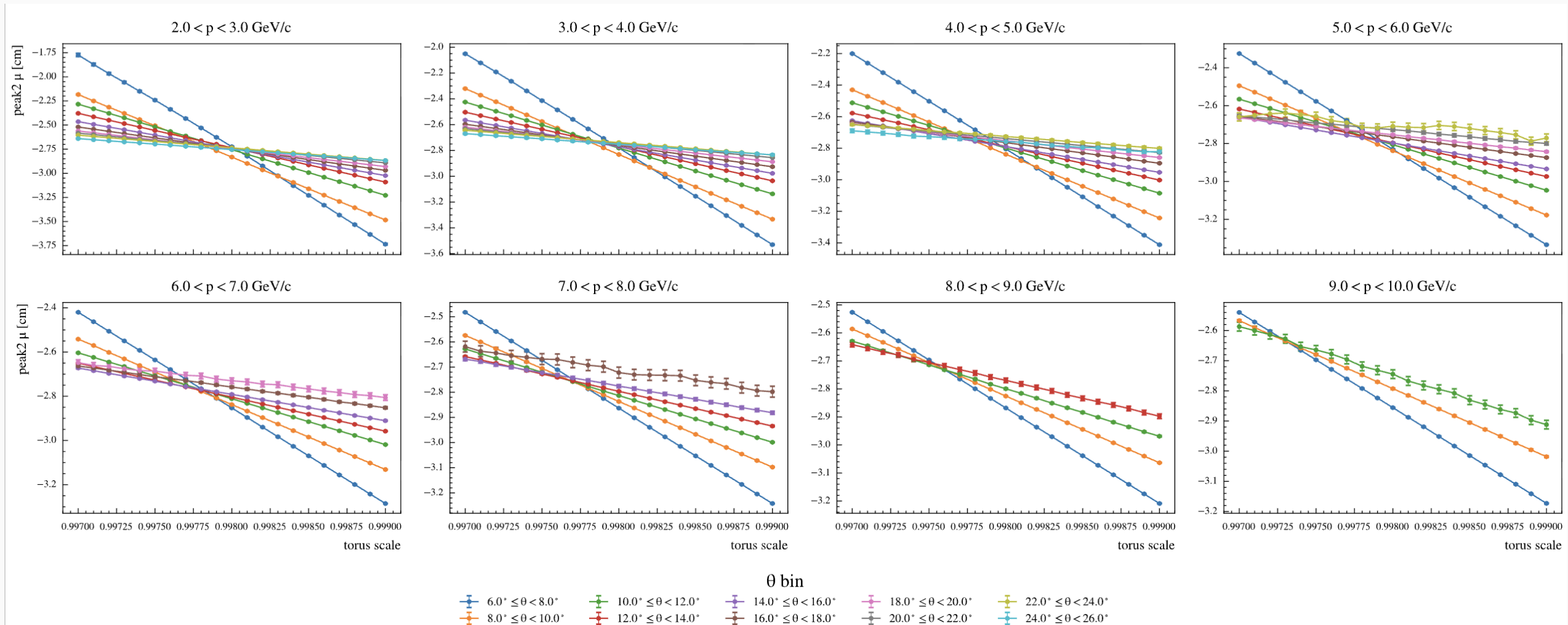
PROBLEM WITH V_z DISTRIBUTION

- Try to change the strength/scale of the torus field: torus scale +0.998



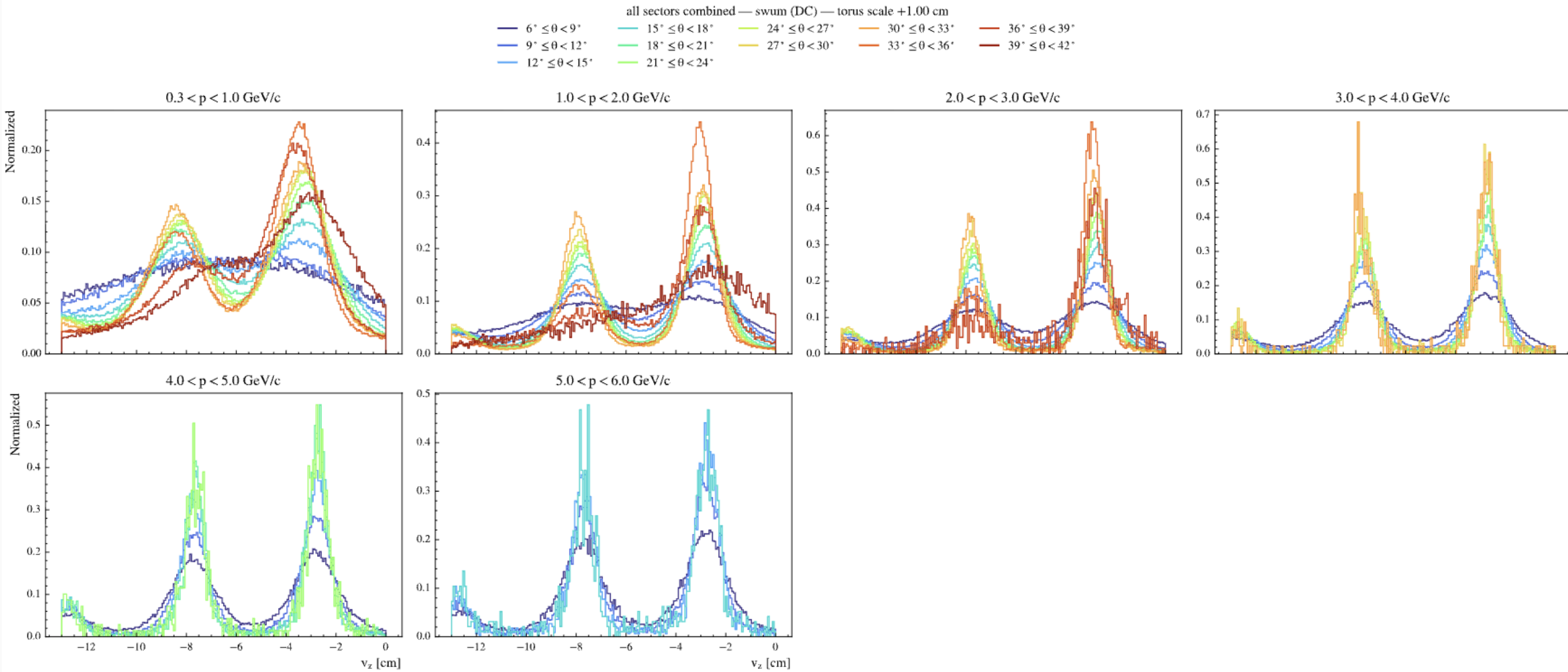
PROBLEM WITH V_Z DISTRIBUTION

- Try to change the strength/scale of the torus field: torus scale +0.998



PROBLEM WITH V_z DISTRIBUTION

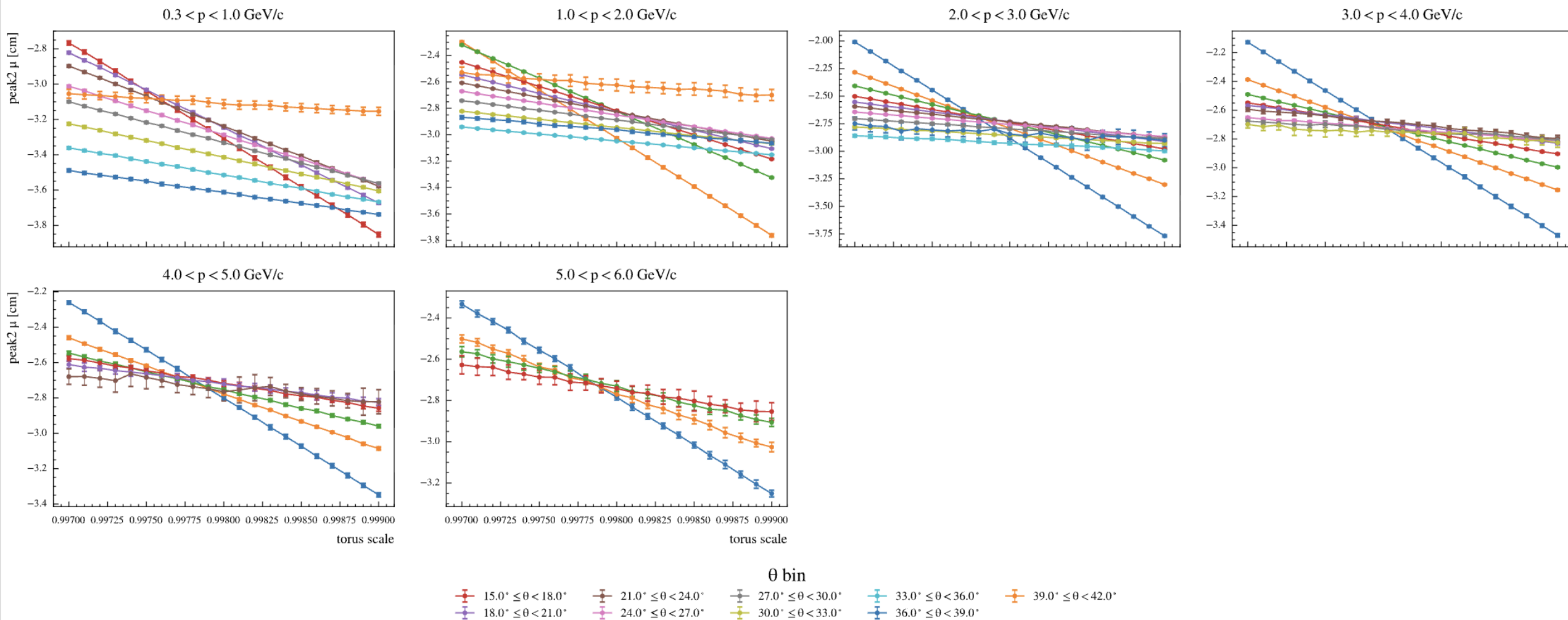
- Test same value for π^- :



PROBLEM WITH V_z DISTRIBUTION

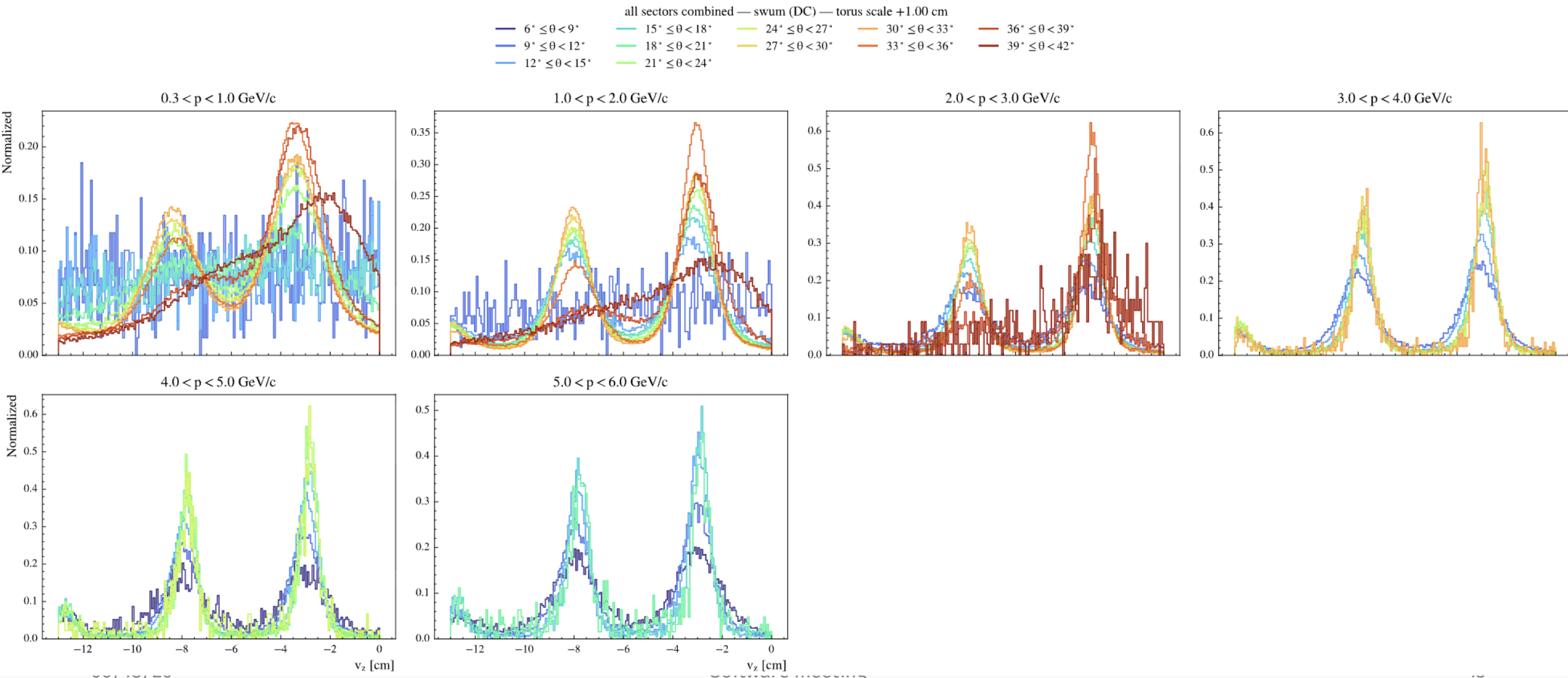
- Test same value for π^+ :

π^- — 2nd-peak mean vs torus scale, per (p, θ)



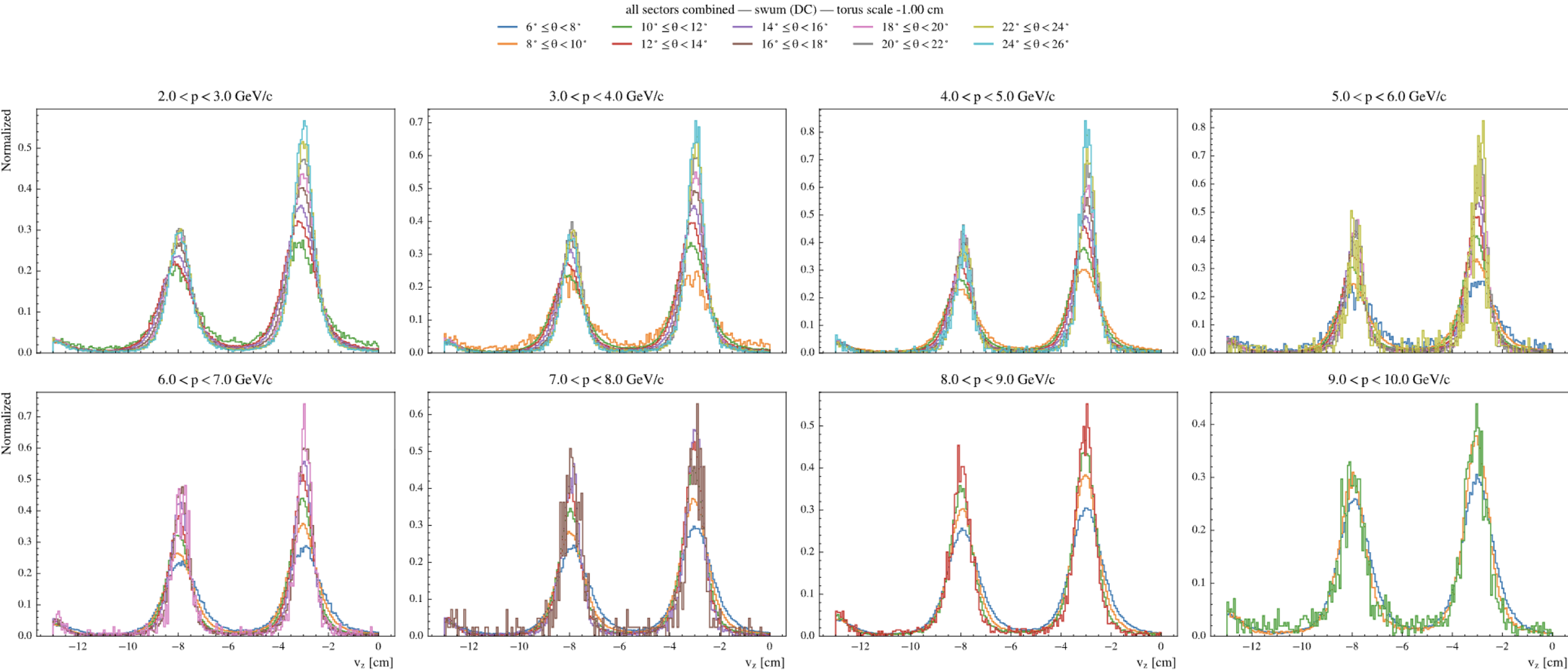
PROBLEM WITH V_z DISTRIBUTION

- Test same value for π^+ :



PROBLEM WITH V_z DISTRIBUTION

- Try also the same scale for electron in inbending configuration



- So, the position of the peak of the V_z distribution shift with momentum and θ
 - Try different modifications of the field maps to fix it
 - Reduce the strength of the torus fields by 0.2% make the shift "dispear"
 - Same effect for electron, negative and positive pions
 - For both outbending and inbending
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- Need to confirm that using the COATJAVA reconstruction, and simulation